

Understanding Latency Sensitivity in Thermal and Tactile Feedback for Multimodal Haptics in VR

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Abstract

Low-latency multimodal feedback is essential for maintaining a high-quality user experience in VR; however, unpredictable network conditions can introduce latency that negatively impacts user experience. This work investigates how users perceive multimodal haptic feedback—specifically thermal (hot/cold) and tactile stimuli—and how latency in such feedback affects user experience. We first measured users’ response times for thermal, tactile, and combined thermal-tactile stimuli. Subsequently, we conducted a psychophysical study to identify delay thresholds for each modality by examining temporal congruency between visual and haptic cues. We designed a haptic delay network simulator to emulate a realistic network environment. Results highlighted that combined thermal-tactile feedback has higher latency tolerance than thermal-only feedback, indicating that multimodal integration can buffer the negative effects of latency. Using these thresholds, we designed controlled latency conditions and assessed user experience. Based on our findings, we propose design recommendations for haptic data transmission in networked VR systems.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; *User studies*; *Haptic devices*; • **Computing methodologies** → **Perception**.

Keywords

Haptics, Thermal Feedback, Tactile Feedback, Multimodal Latency Tolerance, Visual-Haptic Congruency

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1 Introduction

Low-latency feedback is critical to maintaining immersion and realism in virtual reality (VR) systems [19, 27]. Even subtle delays between a user’s action and the system’s response can disrupt the sense of presence, increase cognitive load, and diminish the overall experience. These challenges are especially pronounced in networked VR environments, where fluctuating network conditions can introduce unpredictable latency, making it difficult to synchronize visual, auditory, and haptic cues in real time.

Among various modalities, haptic feedback, particularly tactile and thermal sensations, plays a pivotal role in enhancing realism and embodiment. Previous studies have shown that integrating thermal and tactile cues can significantly enhance realism and immersiveness [44, 51]. Similarly, combining these stimuli in mid-air haptic scenarios using ultrasound arrays has demonstrated comparable benefits [45]. However, tactile and thermal stimuli differ substantially in how they are perceived and processed by the human nervous system. Tactile feedback is rapidly transmitted by mechanoreceptors in the skin, reaching the brain within 20–60 ms [18]. In contrast, thermal feedback—detected by cold and warm receptors in the skin—has significantly slower conduction speeds, with reaction times ranging from 140–200 ms (cold) [32] to over 280–356 ms (warm) [8]. These differences are also reflected in reaction times. Tactile stimuli on the fingertip elicit responses in about 182 ms [30], while cold stimuli evoke reactions around 529 ms, and warm stimuli even slower at approximately 938 ms [16]. These inherent physiological differences create temporal mismatches that can make it difficult to deliver coherent multimodal haptic feedback.

Despite their differing conduction speeds and physiological pathways, the human brain exhibits a remarkable capacity to bind asynchronous sensory inputs into a unified perceptual experience. Notably, prior research has demonstrated that users can perceive thermal and tactile stimuli as temporally congruent even when they are separated by several hundred milliseconds—with simultaneity windows extending up to 1041 ms [20, 21]. This tolerance is thought to arise from cross-modal integration mechanisms and the brain’s adaptive handling of inherently sluggish thermal cues.

Yet, it remains unclear how these perceptual windows behave under dynamic latency conditions typical of networked VR systems. Does the simultaneous delivery of thermal and tactile feedback simply increase perceptual load and make latency more noticeable? Or can multimodal integration act as a buffer, enhancing tolerance

to delay by leveraging redundancy and compensatory mechanisms across modalities?

This paper addresses a key question: Can multimodal haptic integration (tactile + thermal) improve latency tolerance in VR, particularly in scenarios where visual-haptic synchrony is compromised by network-induced delay? To explore this, we conducted two user studies: Study 1 employed a psychophysical method to measure visual-haptic latency thresholds for unimodal (thermal-only or tactile-only) and multimodal (thermal + tactile) feedback conditions. Study 2 evaluated how latency affects user experience—measured via immersion, realism, satisfaction, and presence—across different feedback types in immersive VR scenarios, using a custom-built network latency simulator to closely replicate the behavior of real-world networked environments. Through these studies, we reveal that multimodal haptic feedback significantly improves tolerance to latency compared to unimodal feedback and preserves user experience across a broader range of delay conditions. These findings advance our understanding of multisensory latency perception and offer actionable guidelines for latency-robust haptic design in networked VR systems.

This work makes three key contributions: 1) We establish empirical latency thresholds for thermal, tactile, and combined thermal-tactile stimuli through controlled psychophysical evaluation; 2) We investigated using our network delay simulator how different levels of these latency threshold affects user experience across multiple dimensions including immersion, realism, satisfaction, and presence; and 3) We derive design recommendations for networked VR systems that leverage perceptual integration and cognitive load to reduce the impact of latency on user perception.

2 Related Work

2.1 Unimodal Haptic Latency

Vibrotactile feedback has been widely studied for its effects on perception and richer sensation, but most work is based on zero latency. In practice, vibrotactile latency—the delay between an action (e.g., touching a virtual object) and perceiving feedback, shaped by actuation speed and signal transmission—plays a critical role in information transfer for knowledge communication, data visualization [2–5], immersive authoring [25, 34], and social interactions [23, 24]. Latency plays a critical role in VR interactions, where even small delays in visual stimuli can disrupt collaboration in VR [6, 26]; similar effects are observed with unimodal haptic feedback.

Guidotti et al. [18] explored how fast the somatosensory cortex in the brain processes vibration stimuli, and found the neural transmission latency to be 20–60 ms after actuation. For a visual-haptic feedback supporting AR system developed by Miyatake et al. [33], the vibrotactile haptic latency threshold was 100ms for the index finger, and 500ms for the forearm. Di Luca and Mahnan [11] explored the effect of latency between visual and haptic stimuli, where the participant touched a virtual cube resulting in changing color, and the index finger receiving a vibrotactile stimulation, and reported that the participants did not perceive asynchrony up to 50ms.

Force feedback latency is similar to vibrotactile latency, which involves applying force on a virtual object and the resistive force felt by the subject. Vibrotactile and force feedback actuation times

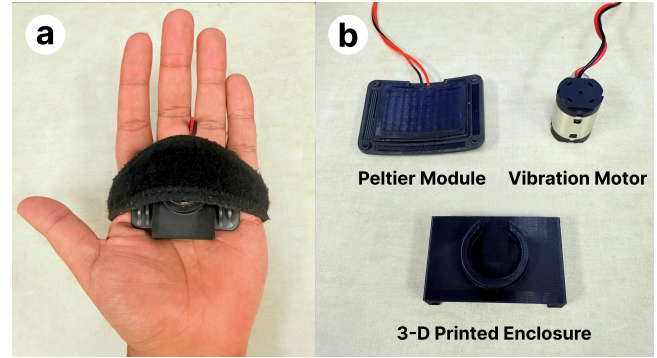


Figure 1: Haptic apparatus. (a) Haptic apparatus placed on the hand with a strap, and (b) peltier module, vibration motor, and the 3D-printed enclosure.

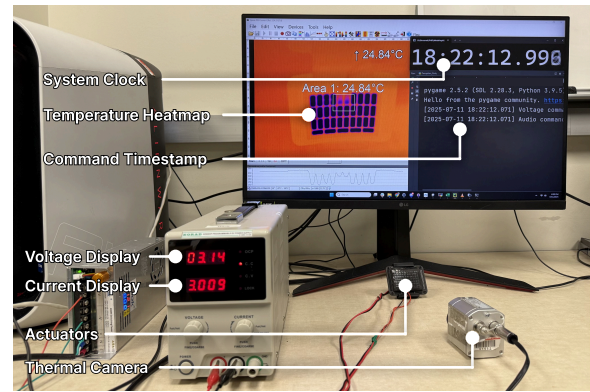


Figure 2: Setup for measuring the system delay of vibrotactile and thermal actuators.

can be affected due to system and network delays. Prior works have simulated network latencies for collaborative tasks and found that latencies below 100 ms do not have a significant impact on task performance [19, 41, 42]. Shi et al. [42] also suggests delaying force actuation 50 ms after the visual cue, as processing visual signals requires more time than processing haptic sensations. Thermal latency refers to the time elapsed when the temperature of the actuator starts changing and when the subject perceives the change. The latency for hot stimuli varies between 280–356 ms, and 140–200 ms for cold stimuli [8, 32].

It can be inferred that non-simultaneity is easily perceivable in the case of vibrotactile sensations compared to thermal sensations. The vibrotactile and force sensations have a lower threshold of around 40–60 ms compared to thermal sensations that vary between 140–529 ms for cold, and 280–938 ms for hot. These works present how each type of haptic feedback is perceived. Our work explores how these unimodal sensations affect qualitative aspects like user experience in immersive environments.

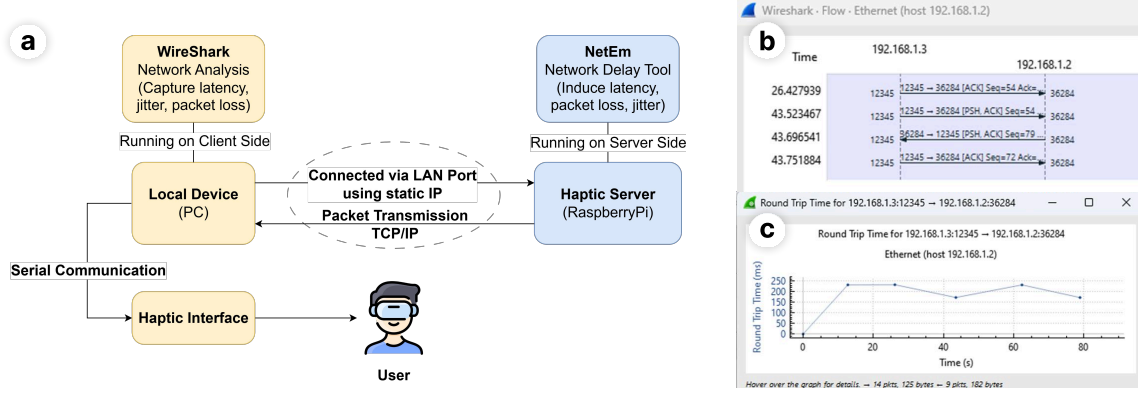


Figure 3: Network Delay Simulator. (a) Connection between the client and haptic server via LAN port, with latency controlled using the NetEm tool and monitored via Wireshark; (b) Analysis of packet transfers between client and server to examine communication; (c) Monitoring latency during data transmission to verify the accuracy of the applied delays.

2.2 Multimodal Haptic Latency

Prior works have explored multisensory feedback to study the effects on perception and develop richer sensations, exploring various parts of the body such as fingers [37, 38, 49], hands [35, 51, 53, 54], arms [52], and torso [46]. Playing multiple stimuli in parallel results in phenomena like phantom thermal sensations on vibrotactile actuators [52] or a higher magnitude of force providing an illusion of hotter or colder temperatures [50]. Our brain combines the various stimuli being perceived in an optimum way based on the prior experiences and assigns higher weightage to the cue with higher resolution [12, 13]. Multimodal feedback incorporates different haptic sensations that are perceived differently and these sensations also use different actuation types which can introduce factors such as latency and asynchrony between them [15].

Audio, video, and haptic multimodal feedback is prevalent across devices like smartphones and video game consoles where haptic feedback, generally in the form of vibrotactile sensation, is delivered in conjunction with an audio or a visual cue. Silva et al. [43] conducted studies to evaluate human perception towards asynchrony in haptic-video and haptic-audio scenarios, found that the participants did not perceive the haptic latency up to 125 ms after the video was played, and 92 ms after the audio was played. Brahimaj et al. [7] suggested to maintain latency below 40 ms for audio-haptic modalities where audio is played first. However, if the haptic are rendered before the audio cue, then the latency should be below 109 ms. In a quality of experience experiment for a haptic media, sound, and video transfer system, addition of 100 ms haptic delay resulted in a lower mean opinion score (MOS) [47]. Thermal sensations in combination with other haptics like vibrotactile feedback can deliver richer sensations like social touch and better perception in VR. In an experiment exploring simultaneity between thermal and tactile cues, it was found that both the cues are perceived to be in synchrony if the warm stimulus precedes tactile by 569 ms, and the cold stimulus precedes by 245 ms [20]. Yi et al. [54] designed a haptic glove that combines thermal and vibro-tactile feedback and they determined that thermal signal had to be activated 242 ms before the tactile cue for simultaneous perception of the stimuli.

While previous studies have investigated both unimodal and multimodal haptic latency, few have examined these effects in immersive VR environments, where spatial presence, cognitive load, and ecological validity are critical. Our work advances this line of research by evaluating latency tolerance across haptic conditions in immersive VR, using network-simulated latency profiles. We also provide new insights into how users perceive the latency of multimodal haptics in VR and offer practical guidelines for minimizing perceptual disruptions in networked VR systems.

3 Setup

To examine latency effects under thermal-vibrotactile conditions in VR, we built an immersive environment with precise control over haptic-visual delay. The system delivers vibrotactile, thermal, and combined cues while simulating network-induced latency. This enables realistic delay profiles for measuring user tolerance. The following describes the system setup and latency control methods.

3.1 Thermal and Vibrotactile Actuators

To deliver multimodal haptic feedback comprising both thermal and vibrotactile sensations, we developed a compact haptic apparatus integrating a Peltier-based thermal actuator and a vibrotactile motor.

Vibration Stimuli. Vibratory feedback was generated using a solenoid-based linear resonant actuator (LRA; Vybronic VG2230001H), driven at 12V via an audio amplifier (see Figure 1-b). A 50 Hz sinusoidal waveform was used to produce a consistent vibration with an acceleration magnitude of 1.029g.

Thermal Stimuli. For thermal stimulation, we employed a Peltier module (Tegway ThermoReal), capable of generating both heating and cooling sensations via thermoelectric regulation (see Figure 1-b). The Peltier was driven at 4V during both hot and cold stimulus conditions. The actuator had a rate of change of +15.3°C during the hot stimulus and -2.6°C during the cold stimulus.

Both actuators were integrated into a custom 3D-printed enclosure that maintained stable skin contact and ensured vibrations were not dampened by thermal components. The design isolated

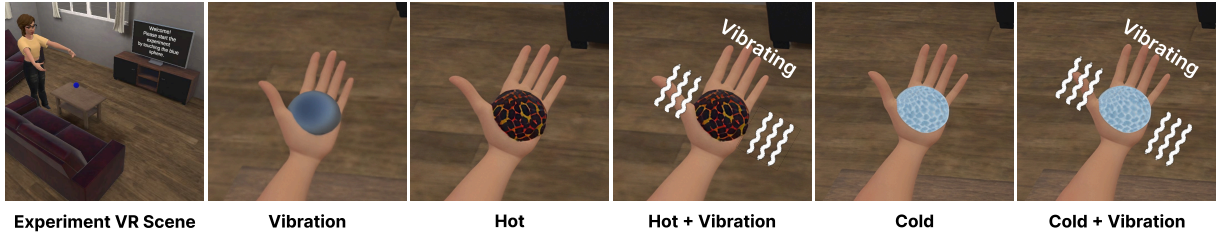


Figure 4: Immersive VR environment and visual cues for all five haptic conditions in the user study 1.

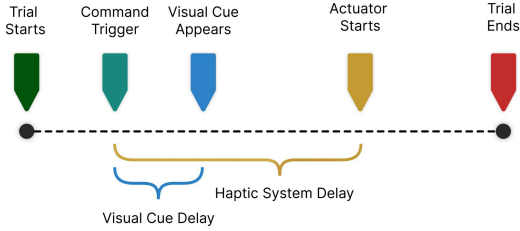


Figure 5: Visual cue and system delays in user study 1.

thermal dissipation and enabled simultaneous, co-located delivery of thermal and vibrotactile stimuli for precise, consistent multi-modal haptic feedback.

3.2 System Delay

Both the vibrotactile and thermal actuators exhibit inherent system delays between the software command and the physical onset of stimulation. To accurately impose artificial latency on haptic stimuli, these baseline delays must be accounted for in the total latency calculation. We developed a custom system clock with millisecond-level (1 ms) accuracy to synchronize actuator commands and annotate timestamps. System delay was calculated by measuring the interval between the PC-issued command and the observable actuation of each haptic device.

To measure the delay of the vibrotactile actuator, we used an iPhone 16 Pro Max to record high-frame-rate slow-motion video at 240 fps, yielding a temporal resolution of 4.17 ms per frame. This enabled frame-by-frame inspection of stimulus onset (see Figure 2). Across five repetitions, the average system delay for the vibrotactile actuator was 88 ms (SD=8.5). For the thermal actuator, we employed a thermal camera (Optris PI OPTPI45IL) to detect the onset of temperature change. The measured delay over five repetitions was 128 ms (SD=4.84).

To report accurate latency thresholds for visual–haptic cue alignment, we incorporated the measured system delays with the artificial network latency (See Figure 5). The visual cue delay, the time between the command trigger and the appearance of the cue in the VR scene, was consistently measured at 26 ms, while the system delays were 88 ms for the vibrotactile actuator and 128 ms for the thermal actuator.

System delays were determined by the limiting bottleneck: 88 ms for vibration, 128 ms for thermal conditions (hot and cold), and 128 ms for multimodal conditions (hot + vibration and cold + vibration), where the thermal actuator served as the bottleneck. These adjusted

delay values were then incorporated into the final latency threshold calculations for each haptic condition.

3.3 Network Delay Simulator

We simulated network-induced haptic latency for vibrotactile and thermal feedback using a dedicated network delay simulator. Rather than embedding fixed delays within application scripts, we introduced latency at the communication layer to better reflect real-world conditions. This approach enabled independent control of latency from the hardware setup, ensuring more consistent management across studies and eliminating potential issues caused by hardware-related variability.

The system consists of a haptics server running on a Raspberry Pi 4, connected to the client PC—running the Unity application—via a direct RJ-45 LAN connection. A client PC was used instead of direct server connection to monitor delays and manage user studies. This setup provides almost negligible network transmission delays, isolating the effects of the artificially introduced latency (see Figure 3-a).

Network delays were emulated using the Linux-based tool *NetEm*¹, which enables precise control over delay, jitter, and packet loss. In our simulator, we controlled only the delay parameter, as introducing jitter or packet loss would add randomness to the network behavior, making the delay conditions harder to reproduce. The setup allowed us to apply latency selectively to specific network interfaces (e.g., Ethernet or Wi-Fi). For this study, we configured both egress and ingress delays on the Ethernet port using standardized *NetEm* commands. We continuously monitored communication latency using *Wireshark*², ensuring that the applied delays matched the target values (see Figure 3-c). This approach allows flexible and precise control of latency without modifying the core application code, enhancing reproducibility across experimental conditions.

4 Study 1: Tolerance for Haptic Latency

The goal of this study was to examine users’ tolerance for haptic latency relative to visual feedback in VR. Specifically, we investigated the perceived temporal congruency between visual and haptic cues to determine the latency threshold, the maximum delay at which the stimuli are still perceived as temporally aligned. Identifying these thresholds offers important insights into defining acceptable latency ranges for networked haptic communication

¹<https://man7.org/linux/man-pages/man8/tc-netem.8.html>

²<https://www.wireshark.org/>

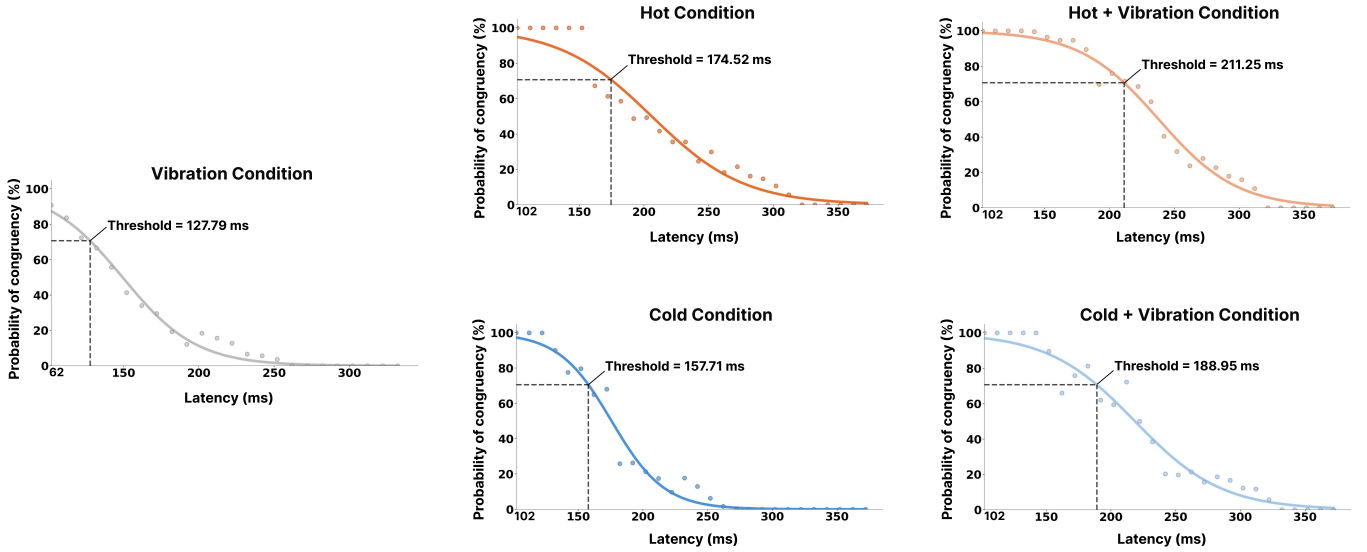


Figure 6: Results of user study 1. Psychophysical thresholds at 70.7% were used to determine delay tolerance across different haptic conditions. These threshold values include the system delay specific to each condition.

systems, where minimizing perceptual asynchrony is critical for maintaining immersive and coherent user experiences.

4.1 Participants

We recruited 16 participants (7 female, 9 male; $M = 24.8$ years, $SD = 3.7$) for the first study. All participants reported no sensory impairments affecting hand sensation. They received a \$10 gift card as compensation. The study was approved by the Institutional Review Board (IRB) of the University of Texas at Dallas (IRB-21-194).

4.2 Study Design

We conducted a within-subject psychophysical experiment to determine the tolerance threshold for visual-haptic latency. The independent variable was the temporal delay of the haptic feedback relative to the visual onset. We employed a 1-up/2-down staircase method with a binary response format to estimate the latency threshold. In each trial, participants were asked whether the visual and haptic cues appeared temporally congruent (“yes”) or mismatched (“no”). This procedure converges to the perceptual threshold to the 70.7% probability point, providing a rigorous estimate of sensory sensitivity [17, 40]. The staircase began at 0 ms latency, with a fixed step size of 10 ms. Each psychophysical experiment continued until eight reversals were recorded, and the average of the last six reversals was used to estimate the individual threshold.

Another independent variable was the type of haptic condition, comprising five levels: vibration, hot, hot + vibration, cold, and cold + vibration. Each condition was visually represented by a spherical cue positioned on the avatar’s left palm. A frosty texture was used to depict the cold condition, while a molten lava texture represented the hot condition. The vibration condition was visualized with a matte metallic sphere that oscillated in place. For the multimodal conditions (cold + vibration and hot + vibration), the corresponding

temperature textures were retained, and the spheres continuously oscillated throughout each trial to reflect the presence of vibration (see Figure 4). Each haptic condition was tested in a separate staircase block, resulting in five staircase procedures per participant. The order of these five conditions was randomized using a Latin square design to control for potential sequence effects.

To simulate real-world networked latency scenarios, we developed the network delay simulator capable of injecting precise delays into the haptic feedback channel as described in Section 3.3. To ensure stable and minimal baseline latency, the haptic server and client machine were connected via a wired LAN, maintaining an initial communication delay of approximately 0 ms. We used the same haptic apparatus from Section 3 with the same thermal and tactile stimulus (see Figure 1).

4.3 Procedure

Participants were seated at a desk and asked to read the study instructions before signing the consent form. They were then assisted in wearing the haptic apparatus on their dominant hand, which included thermal and vibratory actuators secured to their hands with a strap (see Figure 1-a). Next, they entered a VR scene using a Meta Quest 3 headset connected to a PC via Quest Link. In the VR scene, participants were represented by avatars with virtual hands spatially synchronized to their physical hands, where visual cues were displayed (see Figure 4-a). A virtual screen in the scene presented the experimental instructions.

During the experiment, participants responded verbally with “yes” or “no” to indicate whether the timing of the haptic feedback felt congruent with the visual feedback, and their responses were recorded by the experimenter. Verbal responses were chosen over button presses to reduce interaction demands in VR. A 20-second break was given between trials to allow skin temperature to return to baseline.

4.4 Results and Discussion

The results from the psychophysical experiment demonstrated clear differences in participants' tolerance for visual-haptic latency across various haptic conditions. All reported latency threshold values already include the haptic system delays described in Section 3.2. Participants exhibited the lowest tolerance for latency in the vibration condition (127.79 ms), followed by the cold condition (157.71 ms) and the hot condition (174.52 ms). We have added the system delays in these latencies based on haptic conditions, which are illustrated in Figure 6. These findings align with existing literature [16, 30] on response times, indicating that tactile stimuli are perceived more rapidly and sensitively than thermal stimuli, with cold stimuli eliciting greater sensitivity compared to hot stimuli. Thus, latency tolerance between visual and haptic cues appears closely linked to the perceptual processing speed associated with each type of haptic feedback.

Notably, latency tolerance increased significantly in multimodal haptic conditions. The cold + vibration condition had a higher threshold at 188.95 ms, while the hot + vibration condition showed the highest threshold at 211.25 ms. This enhanced tolerance likely arises from the simultaneous processing of thermal and tactile stimuli, which occurs at slightly different perceptual processing speeds. The increased cognitive load and perceptual complexity from integrating multiple modalities might reduce sensitivity to temporal discrepancies relative to visual feedback, leading to greater acceptance of latency.

To confirm these inferences statistically, we examined the normality of the data using Shapiro-Wilk tests, which indicated that the data were normally distributed within each condition (all $p > 0.05$). Mauchly's test indicated a sphericity violation ($\chi^2(9) = 37.48$, $p < 0.001$), so a Greenhouse-Geisser corrected repeated-measures ANOVA was used. The results showed a significant main effect of haptic condition on latency thresholds, $F(4, 60) = 93.04$, $p < 0.001$, $\eta^2 = 0.42$. Post-hoc tests (Bonferroni) revealed significant differences between hot vs. hot+vibration ($p < 0.01$) and cold vs. cold+vibration ($p < 0.05$).

These findings suggest that multimodal haptic conditions yield higher latency tolerance compared to unimodal (thermal or tactile) conditions, likely due to broader temporal integration windows. This interpretation aligns with previous studies by Jodai et al. [20, 21], who reported that the simultaneity window for thermal and tactile perception can extend up to 1041 ms, with subjective simultaneity judgments for cold stimuli ranging between -569 ms and -245 ms. The wide simultaneity window indicates a naturally higher tolerance for temporal asynchrony between these modalities. Consequently, presenting tactile and thermal stimuli together alongside visual cues leverages this perceptual flexibility, resulting in increased thresholds for tolerating visual-haptic latency. In contrast, thermal conditions, which lack immediate onset cues provided by tactile stimuli, present narrower temporal integration windows and thus lower latency tolerance thresholds.

5 Study 2: User Experience on Haptic Latency

The goal of this study was to investigate how varying levels of latency affect user experience across different haptic conditions.

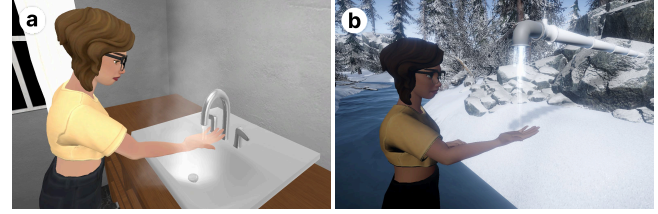


Figure 7: VR scenes in the user study 2. (a) Hot water stream from a tap, and (b) cold water stream from a pipe.

Specifically, we aimed to assess the impact of latency on users' experience in terms of realism, immersion, presence, and satisfaction in both unimodal and multimodal settings.

5.1 Participants

Sixteen participants (7 female, 9 male; $M = 24.5$ years, $SD = 3.4$) completed this study. All participants reported no known sensory impairments affecting hand sensation. They received a \$10 gift card as compensation for 1 hour study. The study was approved by the Institutional Review Board (IRB) of the University of Texas at Dallas (IRB-21-194).

5.2 Study Design

This study adopted a within-subjects design to investigate the effects of latency on user experience across different haptic feedback modalities. We employed the same vibration and thermal apparatus used in the first study (see Figure 1). To evaluate user experience under the VR contexts, two immersive VR scenes were developed, each simulating either hot or cold water flowing from a virtual faucet (see Figure 7).

Each VR scene was paired with contextually appropriate haptic stimuli. Vibration feedback served as the baseline condition across both scenes. In the hot water scene, participants experienced three haptic conditions: vibration only, hot thermal feedback, and combined hot + vibration feedback. Similarly, in the cold water scene, the three corresponding conditions included vibration only, cold thermal feedback, and cold + vibration feedback—resulting in a total of six distinct haptic conditions.

Latency was systematically manipulated across five levels: no delay (0 ms), $1\times$ the latency tolerance threshold (determined individually for each haptic condition based on Study 1), $1.5\times$, $2\times$, and $2.5\times$ the threshold (see Table 1). A step size of $0.5\times$ was chosen to provide finer resolution in capturing the effects of latency.

To simulate realistic latency conditions while maintaining control over hardware performance, a constant system delay was supplemented with artificial delays corresponding to each latency level. This ensured consistent baseline system behavior while emulating network-induced delay. For each haptic condition, the latency levels were calculated by this formula: *Haptic system delay* + (*Perceptual threshold from user study 1* $\times$ *latency level*). For example, in the vibration condition, the $1\times$ threshold latency was calculated as 88 ms (haptic system delay) + 1×39.79 ms (perceptual threshold) = 127.79 ms.

This process was repeated for all haptic conditions using the system delays in Section 3.2, with final latency levels summarized

Table 1: Latency levels across the haptic conditions in the user study 2. Each value was calculated by combining the specific haptic system delay with the artificial delay perceptual threshold obtained from user study 1, scaled by the latency multiplier

	Vibration	Hot	Hot + Vibration	Cold	Cold + Vibration
No Latency	88 + 0 ms	128 + 0 ms	128 + 0 ms	128 + 0 ms	128 + 0 ms
1× Threshold	88 + 39.79 ms	128 + 46.52 ms	128 + 83.25 ms	128 + 29.71 ms	128 + 60.95 ms
1.5× Threshold	88 + 59.68 ms	128 + 69.78 ms	128 + 124.87 ms	128 + 44.56 ms	128 + 91.42 ms
2× Threshold	88 + 79.58 ms	128 + 93.04 ms	128 + 166.5 ms	128 + 59.42 ms	128 + 121.9 ms
2.5× Threshold	88 + 99.47 ms	128 + 116.3 ms	128 + 208.125 ms	128 + 74.27 ms	128 + 152.37 ms

Table 2: A questionnaire of user experience in the user study 2.

	Question
Realism	I felt the visual and haptic cues were realistic.
Immersion	I felt fully engaged while experiencing the visual and haptic cues.
Satisfaction	I felt the visual and haptic cues were satisfying.
Presence	I felt like I was present in the virtual environment when I experienced the visual and haptic cues.

in Table 1. Visual delay was excluded since the stream was already flowing and haptic stimuli were triggered only when the hand entered the water.

All six haptic conditions were tested at five latency levels in randomized order, with four repetitions each, totaling 120 trials per participant. After each trial, participants rated realism, immersion, presence, and satisfaction on a 7-point Likert scale (Table 2) using a custom questionnaire suited for the high-repetition design. A 20-second rest was given between trials, and each session lasted about 60 minutes.

5.3 Procedure

The procedure mirrored the first study, including participant instruction, consent, and setup of the haptic apparatus and VR HMD. Using a Meta Quest 3, participants entered a VR scene by a virtual water stream, where each trial began with the water stream activating, with the on-screen instructions (7).

Participants placed their hand under the stream to receive haptic stimuli with a designated latency level and then rated immersion, presence, satisfaction, and realism on a 7-point Likert scale. Responses were recorded by the experimenter. Conditions were counterbalanced with a block design, and 20-second breaks were given between trials.

5.4 Results and Discussion

Latency levels significantly influenced overall user experience across the dimensions of *realism*, *immersion*, *presence*, and *satisfaction* (see Figure 8). In particular, user experience declined notably when latency exceeded twice the tolerance threshold, with the effect being more pronounced under unimodal conditions.

We statistically analyzed the effects of haptic condition and latency level on user experience using a Generalized Linear Mixed Model (GLMM) suited for Likert-scale data due to non-normality ($p < 0.001$ in Shapiro-Wilk test) and non-homogeneity of variances ($p < 0.001$ in Levene's test). We used a logit link function with haptic condition and latency as fixed effects and participant as a

random effect. Post-hoc pairwise comparisons were adjusted using Bonferroni correction.

Realism. Realism ratings significantly differed by both latency level and haptic condition (*both* $p < 0.001$). Generally, unimodal feedback was more prone to degradation from increased latency than multimodal feedback. In the hot water scene, significant realism drops occurred beyond 1.5× thresholds in both vibration and hot conditions (*all* $p < 0.001$). The multimodal hot+vibration condition showed significant changes only at higher latencies between 2 to 2.5× threshold ($p < 0.001$). In the cold water scene, unimodal feedback showed significant realism reductions even at lower latencies between 1 and 1.5× threshold, while the multimodal cold+vibration condition showed significant differences only at higher latencies between 2 and 2.5× threshold ($p < 0.001$). Cross-haptic condition comparisons indicated that multimodal feedback, with superior latency tolerance, maintained higher realism than unimodal feedback in significant differences between hot and hot+vibration conditions at 2× threshold ($p < 0.05$), and between cold and cold+vibration conditions at 1.5 and 2× thresholds (*all* $p < 0.05$).

Immersion. Immersion ratings were significantly influenced by both latency level and haptic condition (*both* $p < 0.001$). Immersion declined more sharply under unimodal conditions, particularly beyond 1.5× threshold latency (*all* $p < 0.05$). Multimodal conditions showed resilience, with significant declines appearing only at higher latency intervals between 2 and 2.5× threshold (*all* $p < 0.01$). Cross-haptic condition comparisons consistently revealed that multimodal feedback sustained higher immersion levels compared to unimodal feedback in significant differences between hot and hot + vibration at 2× threshold ($p < 0.05$) and between cold and cold + vibration at 1.5 and 2× thresholds (*all* $p < 0.01$).

Presence. Presence ratings significantly varied with both latency level and haptic condition (*both* $p < 0.001$), with multimodal feedback maintaining higher presence ratings up to the 2× threshold. In contrast, unimodal conditions showed earlier and more pronounced declines—beyond 2× threshold in the hot scene and beyond 1.5× threshold in the cold scene (*all* $p < 0.001$). Multimodal feedback

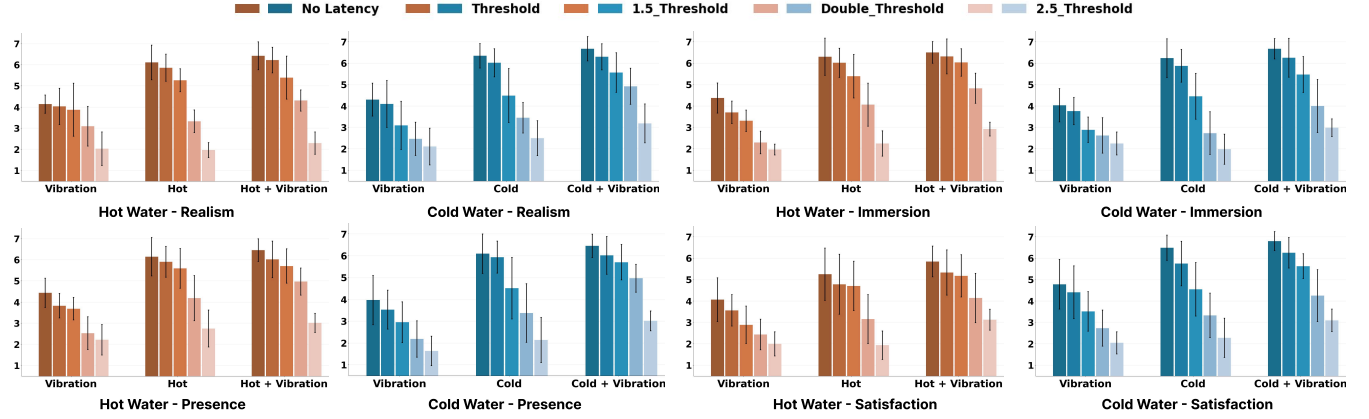


Figure 8: Results of user study 2. Impact of different latency levels on realism, satisfaction, immersion, and presence across different haptic conditions.

only exhibited significant drops at the highest latency level between 2 and 2.5 $\times$ threshold, (*all* $p < 0.001$). Comparisons across haptic conditions further confirmed the superior resilience of multimodal feedback at higher latencies, particularly at 2 and 2.5 $\times$ thresholds in the hot scene, and at 1.5 and 2 $\times$ thresholds in the cold scene.

Satisfaction. Satisfaction ratings were significantly impacted by both latency level and haptic condition (*both* $p < 0.001$). Unimodal conditions showed notable declines in satisfaction beyond 1.5 $\times$ and 1 $\times$ thresholds in the hot and cold scenes, respectively (*all* $p < 0.05$). In contrast, multimodal feedback maintained higher satisfaction ratings up to at least 1.5 $\times$ threshold in both hot + vibration and cold + vibration conditions (*all* $p < 0.05$). Cross-condition comparisons further highlighted the benefits of multimodal feedback. In the hot scene, satisfaction ratings significantly differed between hot and hot + vibration at 2 $\times$ ($p < 0.05$) and 2.5 $\times$ threshold (*all* $p < 0.01$). Similarly, in the cold scene, significant differences were observed between cold and cold + vibration conditions at 1.5 $\times$ and 2.5 $\times$ thresholds (*all* $p < 0.05$).

Overall, across realism, immersion, presence, and satisfaction, multimodal haptics consistently demonstrated superior latency tolerance compared to unimodal conditions, particularly beyond 1.5 $\times$ threshold latency. Multimodal conditions effectively preserved user experiences at higher latency levels, whereas unimodal feedback quickly deteriorated. Additionally, latency impacts were notably stronger in the cold water scene, suggesting differential sensitivity based on thermal modality. These findings highlight the robustness of multimodal haptics in maintaining overall interaction quality under increased latencies.

6 Discussion

This study investigated tolerance thresholds for haptic latency and their effects on user experience in VR, emphasizing the differential impact of unimodal and multimodal haptic feedback. The first user study utilized a psychophysical approach to establish latency tolerance thresholds based on users' perceptions of temporal congruency between visual and haptic cues. Results revealed that multimodal feedback (combining vibration and thermal stimuli) significantly

increased latency tolerance compared to unimodal feedback, highlighting the benefits of multisensory integration in mitigating perceptual disruptions caused by latency. Building upon these thresholds, the second study examined the systematic influence of varied latency levels on four dimensions of user experience: immersion, realism, satisfaction, and presence.

6.1 Insights

Our findings showed that multimodal haptic feedback significantly enhanced latency tolerance compared to unimodal conditions, aligning with established theories of multisensory integration [55]. According to these theories, combining multiple sensory modalities broadens the temporal integration window, thereby reducing perceptual sensitivity to temporal discrepancies [55]. Specifically, integrating thermal (cold or hot) stimuli with tactile vibration yielded substantially higher latency tolerance thresholds. Within the VR context, this highlights the potential of multimodal haptics to preserve perceptual coherence despite latency-induced disruptions. Future studies should explore additional multimodal combinations, including auditory feedback, to further strengthen the robustness of multisensory integration against latency effects.

Several neural and perceptual mechanisms further explain this enhanced latency tolerance in multimodal contexts. Different sensory modalities exhibit inherently distinct neural processing speeds. Visual stimuli typically trigger rapid cortical responses within approximately 100–150 ms, primarily engaging visual cortical regions [39]. Tactile stimuli involve moderately slower neural pathways (approximately 150–200 ms) through the somatosensory cortex [18], while thermal stimuli are processed even more slowly (300–1000 ms), involving the insular cortex and secondary somatosensory areas [48]. Integrating sensory inputs with varied inherent latencies effectively broadens the temporal binding window (TBW)—the period during which asynchronous signals are perceived as simultaneous events—thus reducing sensitivity to small temporal misalignments [1].

Moreover, faster modalities such as tactile vibration can perceptually mask delays in slower modalities like thermal feedback, a phenomenon known as cross-modal compensation [21]. This occurs

as early-arriving signals anchor the perception of the event, diminishing the perceptual impact of later-arriving stimuli. Users also naturally adjust their expectations based on each modality's characteristics; because thermal feedback inherently involves slower physiological transduction, users may implicitly tolerate higher latency when thermal cues are paired with quicker tactile inputs [22].

Increased latency reduced realism, immersion, presence, and satisfaction, but multimodal feedback proved more resilient, maintaining quality even at twice the unimodal latency. This resilience likely stems from the cognitive load of integrating multiple sensory streams, which adds perceptual noise that masks latency effects [14, 36]. Further research into the cognitive and neural mechanisms underlying this phenomenon could provide deeper insights into designing robust multisensory experiences.

Additionally, latency-induced degradation was notably more pronounced for cold feedback compared to hot feedback across all measured experience dimensions. This modality-specific difference suggests narrower temporal integration windows or heightened perceptual sensitivity for cold stimuli [28, 52]. Therefore, latency management strategies in VR systems should particularly prioritize reducing latency for interactions involving cold thermal stimuli to maintain optimal user experiences.

6.2 Implications

Our findings offer valuable guidelines for designing effective haptic experiences in latency-sensitive VR environments. Primarily, multimodal haptic feedback not only enriches sensory immersion but also significantly enhances system robustness against inevitable network delays. By layering rapid tactile feedback (e.g., vibration) with slower modalities (e.g., thermal stimuli), system designers can effectively mask latency, preserving perceptual coherence and overall user experience even in delayed interaction contexts.

Additionally, the user's perceptual context significantly influences latency perception. Adaptive feedback strategies that dynamically adjust based on cognitive load, task complexity, or environmental richness can optimize user experiences. Complex or cognitively demanding scenarios naturally introduce perceptual noise that mitigates sensitivity to latency [14]. Conversely, simpler and less demanding tasks may require stricter latency control. Thus, context-aware adaptive feedback strategies should be considered for the immersive system design, allowing for tailored latency management based on situational demands.

Lastly, the study shows the potential for predictive modeling in latency management. By understanding how different sensory modalities interact and influence user perception, designers can employ predictive models to preemptively adjust haptic feedback timing and intensity, further reducing perceived latency impacts. Such intelligent adaptive systems could dynamically calibrate sensory feedback in real-time, enhancing immersion, realism, and user satisfaction across various latency-prone scenarios.

Overall, incorporating multimodal haptic feedback is essential for managing latency-induced disruptions, particularly in networked VR applications. Further studies exploring sensory modalities, varied latency conditions, individual user differences, and adaptive predictive systems could significantly enrich our understanding of temporal perception and enhance multisensory integration.

6.3 Limitations & Future Work

While this study offers valuable insights, several limitations warrant consideration. First, it focused solely on vibration and thermal feedback, excluding other sensory modalities such as auditory or olfactory cues [31]. Future research should explore broader multi-sensory combinations to broaden the scope of integration strategies.

Second, the controlled latency conditions do not reflect the variability of real-world network environments characterized by unpredictable or fluctuating delays. Investigating user responses under dynamic latency will improve ecological validity.

Third, our study examined a common VR haptic scenario where users first engaged with visual events, followed by real-time haptic feedback. We chose this setup to assess potential perceptual latency effects on user experience. However, the approach can also extend to cases where interactions are predefined and haptic cues are delivered in advance to synchronize multimodal feedback. This alternative, where haptics precede visuals, is equally important for understanding perceptual synchronization. Prior work on multimodal integration shows that effective synchronization can occur even when haptic stimuli precede visual feedback [9, 20, 21, 29].

In addition, individual differences—such as sensory sensitivity, VR experience, or attention capacity—were not considered but may influence latency perception. Incorporating physiological measures (e.g., EEG, EDA) could provide deeper insights into perceptual responses [10]. Finally, future work should develop and validate adaptive systems that dynamically adjust haptic feedback in real time, particularly for high-stakes VR applications such as training simulations, remote collaborations, and virtual therapies.

7 Conclusion

In the first study, we measured latency tolerance across unimodal (hot, cold, vibration) and multimodal (hot + vibration, cold + vibration) haptic feedback. Results showed that multimodal feedback allowed significantly higher tolerance than unimodal feedback, suggesting its effectiveness in mitigating perceptual disruptions under latency. Building on these findings, the second study applied varying latency levels—based on the thresholds identified in Study 1—to assess their impact on user experience, including realism, immersion, presence, and satisfaction. Multimodal haptics consistently demonstrated greater resilience, particularly when latency exceeded $1.5\times$ the threshold, preserving user experience even as unimodal feedback performance degraded. Together, these results highlight the value of multimodal haptics in high-latency scenarios, as combining multiple sensory modalities broadens the temporal integration window and reduces perceptual sensitivity to temporal misalignment.

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